(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

EIGHTH SEMESTER [B.TECH] JUNE 2024

Paper Code: ETIT-402 ***Subject: Mobile Computing***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data if any.

Q1 Attempt **any Five** of the following questions:- **[5x5=25]**

a) What were the reason behind the implementation of WAP services with WML?
b) Mention the features of mobile internet.
c) Explain the WAP architecture in brief.
d) Explain the Data dissemination issues in mobile networks.
e) Differentiate between tunneling and reverse tunneling.
f) With the help of neat protocol stack, draw and explain three-tier architecture for mobile computing. What are mobile nodes? Explain.

UNIT–I

Q2 What is mobile computing? List its key characteristics. Write down the applications of mobile computing. **[12.5]**

Q3 a) Explain Generation of Mobile services. **[6]**
b) Describe the working of a contention-based MAC protocol with suitable examples. **[6.5]**

UNIT–II

Q4 Compare and Explain IEEE 802.11 (WLAN) and IEEE 802.15 (Bluetooth). **[12.5]**

Q5 Write down Short note on following (**any two**): **[6.25, 6.25]**

a) WiMax technology and its application.
b) ZigBee Technology and its usage.
c) Mobile Handover Procedure

UNIT–III

Q6 a) Differentiate between ad-hoc network and infrastructure-based network. **[6]**
b) Explain DSR routing protocol in details. **[6.5]**

Q7 a) What is snooping TCP approach in mobile wireless networks? Discuss its advantages. **[6]**
b) Explain location management in mobile networks. **[6.5]**

UNIT–IV

Q8 a) What are the various methods for Data Synchronization? Explain. **[6]**
b) Explain the concept of wireless markup language. **[6.5]**

Q9 a) Compare the feature sets of Android, Symbian, and Windows Phone operating systems. **[6]**
b) Explain the architecture palm OS. What are the constraints of mobile device OS? **[6.5]**

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ETIT-402

(Please write your Exam Roll No.) *Exam Roll No. .....*

END TERM EXAMINATION

SIXTH SEMESTER [B.TECH] JULY-2023

Paper Code: ETCS-304 ***Subject: Operating Systems***

Time: 3 Hours ***Maximum Marks:75***

Note: Attempt five questions in all including.Qno.1 which is compulsory. Select one question from each unit.

Q1 **Attempt any Five** **[5x5=25]**

(a) What is the meaning of the term *busy waiting*? What other kinds of waiting are there in an operating system? Can busy waiting be avoided altogether? Explain your answer.

(b) What is multiprogramming operating system? How it is different from Multiprocessing. Explain.

(c) Explain Process Control Block with the help of suitable diagram in detail.

(d) Explain race condition with suitable example?

(e) What are the four conditions that must be present for a deadlock to be possible?

(f) What are various file attributes?

UNIT-I

Q2 (a) Explain the layered approach of Operating System? What are main task of OS? **[6]**

(b) Given Memory Partitions of 100K,500K,200K,300K and 600K (in order), how would each of the first fit, best fit, and worst fit algorithms place processes of 212K,417K,112K and 426K (in order)? Which algorithms make the most efficient use of memory? **[6.5]**

Q3 (a) Explain the difference between logical and physical addresses. **[3]**

(b) Why are pages sizes always power of 2? **[3]**

(c) Consider there are 4 frames allocated to a process and the page reference string is:

7,0,1,2,0,3,0,4,2,3,0,3,2,1,2,0,1,7,0,1

Calculate the number of page faults for the FIFO and LRU page replacement algorithms. **[6.5]**

UNIT-II

Q4 (a) Consider the following set of processes with the CPU burst & arrival time in milliseconds. **[6.5]**

Process	Arrival Time	Burst Time
A	0	12
B	1	7
C	2	9
D	3	4
E	4	11

Draw the Gantt chart & find:

P.T.O.

(Please write your Roll No. immediately) **Roll No.**_____

Mid Term Examination

B.Tech (CSE/IT)- 6th Semester **April 2022**

Paper Code:ETCS-304 **Subject: Operating Systems**

Time: 1 ½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q.1. **Answer all the following questions briefly:** **[2 X 5 = 10]**

a) Compare Parallel Systems and distributed Systems.

b) Justify the statement "OS works as a resource manager".

c) Differentiate between Preemptive and non-Preemptive scheduling.

d) Calculate the average amount of internal fragmentation paging scheme may suffer w.r.t to page size.

e) Draw process state diagram.

Q.2.

a) Define interrupts. Why interrupts are important for the functioning of the system. **[5]**

b) In a byte addressable system, the logical address space is of 24 bit where as physical address space is 16 bit. The frame size is 1 KB and each entry in page table requires 2 bytes. Find (i) No. of frames (ii) No. of pages (iii) No. of pages required to store the page table.

Q.3.

a) Differentiate paging and segmentation with the help of neat diagrams. **[5]**

b) Consider a system with following data **[5]**

Process	Burst time in ms	Arrival Time in ms
P0	6	0
P1	2	2
P2	1	4
P3	2	5

Calculate average waiting time and turn around time for SJF and SRTF algorithms.

Q.4.

a) Define following (i) PCB (ii) Context switching (iii) Swapping (iv) Thread (v) IPC **[5]**

b) Assume 3 frames available. The page reference string is 1,2,1,3,2,1,4,5,2,3,1,6,5,4,3,2,1. Calculate no. of page faults in case of FIFO and LRU page replacement algorithms in case of pure demand paging. **[5]**

*******ALL THE BEST*******